

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
INT428	ARTIFICIAL INTELLIGENCE ESSENTIALS	3	0	1	4	
Course Weightage	ATT: 5 CA: 45 ETT: 50					
Course Focus	EMPLOYABILITY,ENTREPRENEURSHIP,SKILL DEVELOPMENT					

Course Outcomes :Through this course students should be able to

CO1 :: Explain the foundations, characteristics, evolution, and applications of Artificial Intelligence across various domains.

CO2 :: Apply state-space modeling and search techniques for solving AI-based problems efficiently.

CO3 :: Apply machine learning algorithms, model evaluation methods, and probabilistic reasoning techniques to AI problems.

CO4 :: Use deep learning and NLP models to design intelligent language-based applications.

CO5 :: Explain generative AI concepts and techniques using LLMs and prompt engineering, while addressing ethical and responsible AI use.

CO6 :: Apply data and AI workflows to use appropriate tools for visualization, model deployment, troubleshooting, and automation using AI and MLOps.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	ARTIFICIAL INTELLIGENCE AND INTELLIGENT SYSTEM	N. P. PADHY	OXFORD UNIVERSITY PRESS

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	ARTIFICIAL INTELLIGENCE: A MODERN APPROACH	STUART RUSSELL, PETER NORVIG	PEARSON
R-2	A FIRST COURSE IN ARTIFICIAL INTELLIGENCE	DEEPAK KHEMANI	MCGRAW HILL EDUCATION

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://onlinecourses.nptel.ac.in/noc25_cs05/preview	AI:Constraint Satisfaction
RW-2	https://onlinecourses.nptel.ac.in/noc20_cs29/preview?	Introduction to Machine Learning
RW-3	https://www.coursera.org/specializations/prompt-engineering	Learn about Prompt Engineering

Audio Visual Aids (AV)		
Sr No	(AV aids) (only if relevant to the course)	Salient Features
AV-1	https://onlinecourses.nptel.ac.in/noc25_cs159/preview , https://elearn.nptel.ac.in/shop/nptel/fundamentals-of-artificial-intelligence/?v=13b5bfe96f3e	Artificial Intelligence: Concepts and Techniques
AV-2	https://onlinecourses.nptel.ac.in/noc24_cs88/preview	Artificial Intelligence: Search Methods for Problem Solving
AV-3	https://onlinecourses.nptel.ac.in/noc24_ma61/preview	Mathematics for Machine Learning
AV-4	https://onlinecourses.nptel.ac.in/noc26_cs33/preview	Deep Learning for Natural Language Processing
AV-5	https://onlinecourses.nptel.ac.in/noc25_cs45/preview	Introduction to Large Language Models (LLMs)
AV-6	https://onlinecourses.nptel.ac.in/noc25_cs118/preview	Responsible & Safe AI Systems
AV-7	https://onlinecourses.nptel.ac.in/noc21_cs45/preview	Data Analytics with Python

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Foundations & Applications of AI(What is Intelligence)	T-1 R-1	AV-1	Lecture 1 will be utilized to discuss zero lecture.	Students will learn about artificial intelligence, its characteristics, and its foundation	Power point presentation, Demonstration using images and videos, class room discussion	Chatbots, smart cars, fraud detection
		Foundations & Applications of AI(what is AI)	T-1 R-1	AV-1	Lecture 1 will be utilized to discuss zero lecture.	Students will learn about artificial intelligence, its characteristics, and its foundation	Power point presentation, Demonstration using images and videos, class room discussion	Chatbots, smart cars, fraud detection

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 1	Lecture 1	Foundations & Applications of AI(characteristics of artificial intelligence)	T-1 R-1	AV-1	Lecture 1 will be utilized to discuss zero lecture.	Students will learn about artificial intelligence, its characteristics, and its foundation	Power point presentation, Demonstration using images and videos, class room discussion	Chatbots, smart cars, fraud detection
		Foundations & Applications of AI(Foundations of AI)	T-1 R-1	AV-1	Lecture 1 will be utilized to discuss zero lecture.	Students will learn about artificial intelligence, its characteristics, and its foundation	Power point presentation, Demonstration using images and videos, class room discussion	Chatbots, smart cars, fraud detection
	Lecture 2	Foundations & Applications of AI(Evolution, and types of AI (narrow, general))	T-1 R-1	AV-1	Lecture 2 will be utilized to discuss the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Students will learn about the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Power point presentation, Demonstration using images and videos, class room discussion	Chatbots, smart cars, fraud detection, Image generation
		Foundations & Applications of AI(Key AI problems and techniques)	T-1 R-1	AV-1	Lecture 2 will be utilized to discuss the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Students will learn about the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Power point presentation, Demonstration using images and videos, class room discussion	Chatbots, smart cars, fraud detection, Image generation
		Foundations & Applications of AI(Applications across domains (business, healthcare, automation, vision, language))	T-1 R-1	AV-1	Lecture 2 will be utilized to discuss the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Students will learn about the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Power point presentation, Demonstration using images and videos, class room discussion	Chatbots, smart cars, fraud detection, Image generation
		Foundations & Applications of AI(Modern AI Toolkits (TensorFlow, PyTorch))	T-1 R-1	AV-1	Lecture 2 will be utilized to discuss the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Students will learn about the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Power point presentation, Demonstration using images and videos, class room discussion	Chatbots, smart cars, fraud detection, Image generation

Week 1	Lecture 2	Foundations & Applications of AI(Responsible AI)	T-1 R-1	AV-1 AV-6	Lecture 2 will be utilized to discuss the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Students will learn about the evolution and types of AI, key AI techniques, applications across domains, modern AI tools, and Responsible AI.	Power point presentation, Demonstration using images and videos, class room discussion	Chatbots, smart cars, fraud detection, Image generation
	Lecture 3	AI Problem Modeling & Search Concepts(Defining AI problems as State Space and Search Problems)	T-1 R-1	AV-1 AV-2	Lecture 3 to 4 will be utilized to discuss AI problem formulation using state space and search, and the characteristics of AI problem spaces.	Students will learn about AI problem formulation using state space and search, and the characteristics of AI problem spaces.	Power point presentation, White board	Water jug problem, Missionaries and Cannibals problem
		AI Problem Modeling & Search Concepts (Characteristics of AI Problem Spaces)	T-1 R-1	AV-1	Lecture 3 to 4 will be utilized to discuss AI problem formulation using state space and search, and the characteristics of AI problem spaces.	Students will learn about AI problem formulation using state space and search, and the characteristics of AI problem spaces.	Power point presentation, White board	Water jug problem, Missionaries and Cannibals problem
Week 2	Lecture 4	AI Problem Modeling & Search Concepts(Defining AI problems as State Space and Search Problems)	T-1 R-1	AV-1 AV-2	Lecture 3 to 4 will be utilized to discuss AI problem formulation using state space and search, and the characteristics of AI problem spaces.	Students will learn about AI problem formulation using state space and search, and the characteristics of AI problem spaces.	Power point presentation, White board	Water jug problem, Missionaries and Cannibals problem
		AI Problem Modeling & Search Concepts (Characteristics of AI Problem Spaces)	T-1 R-1	AV-1	Lecture 3 to 4 will be utilized to discuss AI problem formulation using state space and search, and the characteristics of AI problem spaces.	Students will learn about AI problem formulation using state space and search, and the characteristics of AI problem spaces.	Power point presentation, White board	Water jug problem, Missionaries and Cannibals problem
	Lecture 5	AI Problem Modeling & Search Concepts (Introduction to AI Workflows & Data-Centric Modeling)	R-1 R-2	AV-1	Lecture 5 will be utilized to discuss AI workflows, data-centric modeling, and challenges in AI problem solving.	Students will learn about AI workflows, data-centric modeling, and challenges in AI problem solving.	PowerPoint presentation, Whiteboard explanation, Case study discussion, Classroom interaction	House Price Prediction workflow, Spam Email Classification workflow

Week 2	Lecture 5	AI Problem Modeling & Search Concepts(Challenges in AI problem solving)	R-1 R-2	AV-1	Lecture 5 will be utilized to discuss AI workflows, data-centric modeling, and challenges in AI problem solving.	Students will learn about AI workflows, data-centric modeling, and challenges in AI problem solving.	PowerPoint presentation, Whiteboard explanation, Case study discussion, Classroom interaction	House Price Prediction workflow, Spam Email Classification workflow
	Lecture 6	Problem Solving & Search in AI(Problem formulation and state space search)	T-1 R-1	AV-1	Lecture 6 to Lecture 7 will be utilized to discuss problem formulation, state space, and uninformed search.	Student will learn about problem formulation, state space, and uninformed search.	Power point presentation, White board	Missionaries and Cannibals problem, water jug problem
		Problem Solving & Search in AI(Introduction to uninformed search)	T-1 R-1	AV-1 AV-2	Lecture 6 to Lecture 7 will be utilized to discuss problem formulation, state space, and uninformed search.	Student will learn about problem formulation, state space, and uninformed search.	Power point presentation, White board	Missionaries and Cannibals problem, water jug problem
Week 3	Lecture 7	Problem Solving & Search in AI(Problem formulation and state space search)	T-1 R-1	AV-1	Lecture 6 to Lecture 7 will be utilized to discuss problem formulation, state space, and uninformed search.	Student will learn about problem formulation, state space, and uninformed search.	Power point presentation, White board	Missionaries and Cannibals problem, water jug problem
		Problem Solving & Search in AI(Introduction to uninformed search)	T-1 R-1	AV-1 AV-2	Lecture 6 to Lecture 7 will be utilized to discuss problem formulation, state space, and uninformed search.	Student will learn about problem formulation, state space, and uninformed search.	Power point presentation, White board	Missionaries and Cannibals problem, water jug problem
	Lecture 8	Problem Solving & Search in AI(Core search algorithms: Best-first search, A* search)	T-1 R-1	AV-1	Lecture 8 to Lecture 9 will be utilized to discuss core search algorithms, including Best-First Search, A Search, and Heuristic Search.	Student will learn about core search algorithms, including Best-First Search, A* Search, and Heuristic Search.	Power point presentation, Demonstration using images and videos, class room discussion	8-puzzle problem, Route finding on a map, Maze solving problem
		Problem Solving & Search in AI(Heuristic search)	T-1 R-1	AV-1	Lecture 8 to Lecture 9 will be utilized to discuss core search algorithms, including Best-First Search, A Search, and Heuristic Search.	Student will learn about core search algorithms, including Best-First Search, A* Search, and Heuristic Search.	Power point presentation, Demonstration using images and videos, class room discussion	8-puzzle problem, Route finding on a map, Maze solving problem

Week 3	Lecture 9	Problem Solving & Search in AI(Core search algorithms: Best-first search, A* search)	T-1 R-1	AV-1	Lecture 8 to Lecture 9 will be utilized to discuss core search algorithms, including Best-First Search, A Search, and Heuristic Search.	Student will learn about core search algorithms, including Best-First Search, A* Search, and Heuristic Search.	Power point presentation, Demonstration using images and videos, class room discussion	8-puzzle problem, Route finding on a map, Maze solving problem
		Problem Solving & Search in AI(Heuristic search)	T-1 R-1	AV-1	Lecture 8 to Lecture 9 will be utilized to discuss core search algorithms, including Best-First Search, A Search, and Heuristic Search.	Student will learn about core search algorithms, including Best-First Search, A* Search, and Heuristic Search.	Power point presentation, Demonstration using images and videos, class room discussion	8-puzzle problem, Route finding on a map, Maze solving problem
Week 4	Lecture 10	Problem Solving & Search in AI(Constraint satisfaction)	T-1 R-1	RW-1	Lecture 10 will be utilized to discuss Constraint Satisfaction Problems (CSPs) and their applications.	Student will learn about Constraint Satisfaction Problems (CSPs) and their applications, along with the allocation of Skill-based CA.	PowerPoint presentation, Whiteboard, Classroom interaction	Map Coloring, Sudoku
	Lecture 11	Problem Solving & Search in AI(Basics of optimization: gradient-based methods and metaheuristics)	T-1 R-1		Lecture 11 will be utilized to discuss the basics of optimization, including gradient-based methods and metaheuristic techniques.	Students will learn about the basics of optimization, including gradient-based methods and metaheuristic techniques.	PowerPoint presentation, Whiteboard, Live examples, Classroom interaction	
	Lecture 12	AI problem design (Complexity)	R-1 R-2		Lecture 12 will be utilized to discuss complexity, data requirements, solution metrics, and an introduction to reinforcement learning for sequential decision problems.	Students will learn about complexity, data requirements, solution metrics, and an introduction to reinforcement learning for sequential decision problems.	Power point presentation, Demonstration using images and videos, class room discussion	
		AI problem design(Data requirements)	R-1 R-2		Lecture 12 will be utilized to discuss complexity, data requirements, solution metrics, and an introduction to reinforcement learning for sequential decision problems.	Students will learn about complexity, data requirements, solution metrics, and an introduction to reinforcement learning for sequential decision problems.	Power point presentation, Demonstration using images and videos, class room discussion	

Week 4	Lecture 12	AI problem design(Solution metrics)	R-1 R-2		Lecture 12 will be utilized to discuss complexity, data requirements, solution metrics, and an introduction to reinforcement learning for sequential decision problems.	Students will learn about complexity, data requirements, solution metrics, and an introduction to reinforcement learning for sequential decision problems.	Power point presentation, Demonstration using images and videos, class room discussion	
		AI problem design (Introduction to reinforcement learning for sequential decision problems)	R-1 R-2		Lecture 12 will be utilized to discuss complexity, data requirements, solution metrics, and an introduction to reinforcement learning for sequential decision problems.	Students will learn about complexity, data requirements, solution metrics, and an introduction to reinforcement learning for sequential decision problems.	Power point presentation, Demonstration using images and videos, class room discussion	
Week 5	Lecture 13	Machine Learning (Probability)	T-1 R-1	RW-2 AV-3	Lecture 13 to Lecture 14 will be utilized to discuss probability, statistics, and linear algebra with an applied focus for machine learning.	Students will learn about probability, statistics, and linear algebra with an applied focus for machine learning.	PowerPoint presentation, Whiteboard, Classroom interaction	Dice/coin probability experiments, Descriptive statistics on sample datasets, Matrix operations for ML (dot product, transformations)
		Machine Learning(Statistics)	T-1 R-1	RW-2 AV-3	Lecture 13 to Lecture 14 will be utilized to discuss probability, statistics, and linear algebra with an applied focus for machine learning.	Students will learn about probability, statistics, and linear algebra with an applied focus for machine learning.	PowerPoint presentation, Whiteboard, Classroom interaction	Dice/coin probability experiments, Descriptive statistics on sample datasets, Matrix operations for ML (dot product, transformations)

Week 5	Lecture 13	Machine Learning(Linear algebra (applied focus))	T-1 R-1	RW-2 AV-3	Lecture 13 to Lecture 14 will be utilized to discuss probability, statistics, and linear algebra with an applied focus for machine learning.	Students will learn about probability, statistics, and linear algebra with an applied focus for machine learning.	PowerPoint presentation, Whiteboard, Classroom interaction	Dice/coin probability experiments, Descriptive statistics on sample datasets, Matrix operations for ML (dot product, transformations)
	Lecture 14	Machine Learning (Probability)	T-1 R-1	RW-2 AV-3	Lecture 13 to Lecture 14 will be utilized to discuss probability, statistics, and linear algebra with an applied focus for machine learning.	Students will learn about probability, statistics, and linear algebra with an applied focus for machine learning.	PowerPoint presentation, Whiteboard, Classroom interaction	Dice/coin probability experiments, Descriptive statistics on sample datasets, Matrix operations for ML (dot product, transformations)
		Machine Learning(Statistics)	T-1 R-1	RW-2 AV-3	Lecture 13 to Lecture 14 will be utilized to discuss probability, statistics, and linear algebra with an applied focus for machine learning.	Students will learn about probability, statistics, and linear algebra with an applied focus for machine learning.	PowerPoint presentation, Whiteboard, Classroom interaction	Dice/coin probability experiments, Descriptive statistics on sample datasets, Matrix operations for ML (dot product, transformations)
		Machine Learning(Linear algebra (applied focus))	T-1 R-1	RW-2 AV-3	Lecture 13 to Lecture 14 will be utilized to discuss probability, statistics, and linear algebra with an applied focus for machine learning.	Students will learn about probability, statistics, and linear algebra with an applied focus for machine learning.	PowerPoint presentation, Whiteboard, Classroom interaction	Dice/coin probability experiments, Descriptive statistics on sample datasets, Matrix operations for ML (dot product, transformations)

Week 5	Lecture 15	Machine Learning (Supervised, unsupervised, and reinforcement learning: concepts and real-world use)	R-1 R-2	RW-2	Lecture 15 to Lecture 16 will be utilized to discuss supervised, unsupervised, and reinforcement learning, including their concepts, real-world applications, feature engineering, and model evaluation.	Students will learn about supervised, unsupervised, and reinforcement learning, including their concepts, real-world applications, feature engineering, and model evaluation.	Power point presentation, Demonstration using images and videos, class room discussion	House price prediction (Supervised), Customer segmentation (Unsupervised), Self-driving car simulation (Reinforcement Learning)
		Machine Learning (Feature engineering and model evaluation (cross-validation, precision, recall))	R-1 R-2	RW-2	Lecture 15 to Lecture 16 will be utilized to discuss supervised, unsupervised, and reinforcement learning, including their concepts, real-world applications, feature engineering, and model evaluation.	Students will learn about supervised, unsupervised, and reinforcement learning, including their concepts, real-world applications, feature engineering, and model evaluation.	Power point presentation, Demonstration using images and videos, class room discussion	House price prediction (Supervised), Customer segmentation (Unsupervised), Self-driving car simulation (Reinforcement Learning)
Week 6	Lecture 16	Machine Learning (Supervised, unsupervised, and reinforcement learning: concepts and real-world use)	R-1 R-2	RW-2	Lecture 15 to Lecture 16 will be utilized to discuss supervised, unsupervised, and reinforcement learning, including their concepts, real-world applications, feature engineering, and model evaluation.	Students will learn about supervised, unsupervised, and reinforcement learning, including their concepts, real-world applications, feature engineering, and model evaluation.	Power point presentation, Demonstration using images and videos, class room discussion	House price prediction (Supervised), Customer segmentation (Unsupervised), Self-driving car simulation (Reinforcement Learning)
		Machine Learning (Feature engineering and model evaluation (cross-validation, precision, recall))	R-1 R-2	RW-2	Lecture 15 to Lecture 16 will be utilized to discuss supervised, unsupervised, and reinforcement learning, including their concepts, real-world applications, feature engineering, and model evaluation.	Students will learn about supervised, unsupervised, and reinforcement learning, including their concepts, real-world applications, feature engineering, and model evaluation.	Power point presentation, Demonstration using images and videos, class room discussion	House price prediction (Supervised), Customer segmentation (Unsupervised), Self-driving car simulation (Reinforcement Learning)
	Lecture 17				Test 1			
	Lecture 18	Machine Learning (Bayes theorem)	R-1	RW-2	Lecture 18 to 19 will be utilized to discuss Bayes' Theorem, Bayesian networks, and probabilistic reasoning in machine learning.	Students will learn about Bayes' Theorem, Bayesian networks, and probabilistic reasoning in machine learning.	PowerPoint presentation, Whiteboard	

Week 6	Lecture 18	Machine Learning(Bayesian networks, and probabilistic reasoning)	R-1	RW-2	Lecture 18 to 19 will be utilized to discuss Bayes’ Theorem, Bayesian networks, and probabilistic reasoning in machine learning.	Students will learn about Bayes’ Theorem, Bayesian networks, and probabilistic reasoning in machine learning.	PowerPoint presentation, Whiteboard	
Week 7	Lecture 19	Machine Learning(Bayes theorem)	R-1	RW-2	Lecture 18 to 19 will be utilized to discuss Bayes’ Theorem, Bayesian networks, and probabilistic reasoning in machine learning.	Students will learn about Bayes’ Theorem, Bayesian networks, and probabilistic reasoning in machine learning.	PowerPoint presentation, Whiteboard	
		Machine Learning(Bayesian networks, and probabilistic reasoning)	R-1	RW-2	Lecture 18 to 19 will be utilized to discuss Bayes’ Theorem, Bayesian networks, and probabilistic reasoning in machine learning.	Students will learn about Bayes’ Theorem, Bayesian networks, and probabilistic reasoning in machine learning.	PowerPoint presentation, Whiteboard	
		SPILL OVER						
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
		MID-TERM						
Week 8	Lecture 22	Introduction to deep neural networks(Introduction to Neural Networks)	R-1	AV-4	Lecture 22 will be utilized to discuss the introduction to neural networks and the Perceptron model.	Students will learn about the introduction to neural networks and the Perceptron model.	PowerPoint presentation, Whiteboard	Binary classification using Perceptron
		Introduction to deep neural networks(Perceptron)	R-1	AV-4	Lecture 22 will be utilized to discuss the introduction to neural networks and the Perceptron model.	Students will learn about the introduction to neural networks and the Perceptron model.	PowerPoint presentation, Whiteboard	Binary classification using Perceptron
	Lecture 23	Introduction to deep neural networks(MLP)	R-1		Lecture 23 will be utilized to discuss Multilayer Perceptron (MLP) and their working principles.	Students will learn about Multilayer Perceptron (MLP).	PowerPoint presentation, Whiteboard	

Week 8	Lecture 24	Introduction to deep neural networks(CNN)	R-1		Lecture 23 will be utilized to discuss Convolutional Neural Networks (CNN), their architecture, and applications.	Students will learn about Convolutional Neural Networks (CNN).	PowerPoint presentation, Whiteboard	Image classification, Object recognition in images
Week 9	Lecture 25	Introduction to deep neural networks(RNN)	R-1	AV-4	Lecture 25 will be utilized to discuss Recurrent Neural Networks (RNN), Transformer architecture, and their applications.	Students will learn about Recurrent Neural Networks (RNN), Transformer architecture, and their applications.	Powerpoint presentation, Demonstration using images and videos, class room discussion	Sentiment analysis
		Introduction to deep neural networks(Transformer Architecture and Applications)	R-1	AV-4	Lecture 25 will be utilized to discuss Recurrent Neural Networks (RNN), Transformer architecture, and their applications.	Students will learn about Recurrent Neural Networks (RNN), Transformer architecture, and their applications.	Powerpoint presentation, Demonstration using images and videos, class room discussion	Sentiment analysis
	Lecture 26	Modern NLP(Introduction to NLP)	R-1	AV-4	Lecture 26 will be utilized to introduce Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	Students will learn about Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	PowerPoint presentation, Whiteboard	Text summarization
		Modern NLP(NLP phases)	R-1	AV-4	Lecture 26 will be utilized to introduce Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	Students will learn about Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	PowerPoint presentation, Whiteboard	Text summarization
		Modern NLP(Tokenization)	R-1	AV-4	Lecture 26 will be utilized to introduce Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	Students will learn about Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	PowerPoint presentation, Whiteboard	Text summarization

Week 9	Lecture 26	Modern NLP(Embeddings)	R-1	AV-4	Lecture 26 will be utilized to introduce Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	Students will learn about Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	PowerPoint presentation, Whiteboard	Text summarization
		Modern NLP(Attention)	R-1	AV-4	Lecture 26 will be utilized to introduce Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	Students will learn about Natural Language Processing (NLP), covering its phases, tokenization, embeddings, and attention mechanisms.	PowerPoint presentation, Whiteboard	Text summarization
	Lecture 27	Modern NLP(language models (BERT, GPT))	R-1 R-2	AV-4	Lecture 27 will be utilized to discuss language models (BERT, GPT), building chatbots and digital assistants, and NLP use cases such as sentiment analysis, translation, and summarization.	Students will learn about language models (BERT, GPT), building chatbots and digital assistants, and NLP use cases such as sentiment analysis, translation, and summarization.	Powerpoint presentation, Demonstration using images and videos, class room discussion	alexa, siri, chatbot
		Modern NLP(Building chatbots and digital assistants)	R-1 R-2	AV-4	Lecture 27 will be utilized to discuss language models (BERT, GPT), building chatbots and digital assistants, and NLP use cases such as sentiment analysis, translation, and summarization.	Students will learn about language models (BERT, GPT), building chatbots and digital assistants, and NLP use cases such as sentiment analysis, translation, and summarization.	Powerpoint presentation, Demonstration using images and videos, class room discussion	alexa, siri, chatbot
		Modern NLP(NLP use cases (sentiment analysis, translation, summarization))	R-1 R-2	AV-4	Lecture 27 will be utilized to discuss language models (BERT, GPT), building chatbots and digital assistants, and NLP use cases such as sentiment analysis, translation, and summarization.	Students will learn about language models (BERT, GPT), building chatbots and digital assistants, and NLP use cases such as sentiment analysis, translation, and summarization.	Powerpoint presentation, Demonstration using images and videos, class room discussion	alexa, siri, chatbot

Week 10	Lecture 28	Generative AI & Ethics (Fundamentals of Generative AI)		AV-7	Lecture 28 will be utilized to discuss the fundamentals of Generative AI, including Large Language Models (LLMs), Generative Adversarial Networks (GANs), and Diffusion models.	Students will learn about the fundamentals of Generative AI, including Large Language Models (LLMs), Generative Adversarial Networks (GANs), and Diffusion models.	PowerPoint presentation, Whiteboard	Text generation with LLMs, Image generation using GANs
		Generative AI & Ethics (LLMs)		AV-7	Lecture 28 will be utilized to discuss the fundamentals of Generative AI, including Large Language Models (LLMs), Generative Adversarial Networks (GANs), and Diffusion models.	Students will learn about the fundamentals of Generative AI, including Large Language Models (LLMs), Generative Adversarial Networks (GANs), and Diffusion models.	PowerPoint presentation, Whiteboard	Text generation with LLMs, Image generation using GANs
		Generative AI & Ethics (GANs)		AV-7	Lecture 28 will be utilized to discuss the fundamentals of Generative AI, including Large Language Models (LLMs), Generative Adversarial Networks (GANs), and Diffusion models.	Students will learn about the fundamentals of Generative AI, including Large Language Models (LLMs), Generative Adversarial Networks (GANs), and Diffusion models.	PowerPoint presentation, Whiteboard	Text generation with LLMs, Image generation using GANs
		Generative AI & Ethics (Diffusion)		AV-7	Lecture 28 will be utilized to discuss the fundamentals of Generative AI, including Large Language Models (LLMs), Generative Adversarial Networks (GANs), and Diffusion models.	Students will learn about the fundamentals of Generative AI, including Large Language Models (LLMs), Generative Adversarial Networks (GANs), and Diffusion models.	PowerPoint presentation, Whiteboard	Text generation with LLMs, Image generation using GANs
	Lecture 29	Generative AI & Ethics (How generative models work)		AV-5	Lecture 29 will be utilized to discuss how generative models work, their industrial applications, and the ethics and responsible use of Generative AI.	Students will learn about how generative models work, their industrial applications, and the ethics and responsible use of Generative AI.	Power point presentation, Demonstration using images and videos, class room discussion	Demo of AI-generated content

Week 10	Lecture 29	Generative AI & Ethics (Industrial applications (content creation, automation))		AV-5	Lecture 29 will be utilized to discuss how generative models work, their industrial applications, and the ethics and responsible use of Generative AI.	Students will learn about how generative models work, their industrial applications, and the ethics and responsible use of Generative AI.	Power point presentation, Demonstration using images and videos, class room discussion	Demo of AI-generated content
		Generative AI & Ethics (Ethics and Responsible Use of Generative AI)		AV-5	Lecture 29 will be utilized to discuss how generative models work, their industrial applications, and the ethics and responsible use of Generative AI.	Students will learn about how generative models work, their industrial applications, and the ethics and responsible use of Generative AI.	Power point presentation, Demonstration using images and videos, class room discussion	Demo of AI-generated content
	Lecture 30	Prompt Engineering (Fundamentals of prompt)		RW-3 AV-5	Lecture 30 will be utilized to discuss the fundamentals of prompts, the importance of prompt engineering, an overview of language models, and the key elements of a good prompt.	Students will learn about the fundamentals of prompts, the importance of prompt engineering, an overview of language models, and the key elements of a good prompt.	Powerpoint presentation, Demonstration using images and videos, class room discussion	Bad Prompt vs. Good Prompt Challenge
		Prompt Engineering (Importance of Prompt Engineering)		RW-3 AV-5	Lecture 30 will be utilized to discuss the fundamentals of prompts, the importance of prompt engineering, an overview of language models, and the key elements of a good prompt.	Students will learn about the fundamentals of prompts, the importance of prompt engineering, an overview of language models, and the key elements of a good prompt.	Powerpoint presentation, Demonstration using images and videos, class room discussion	Bad Prompt vs. Good Prompt Challenge
		Prompt Engineering (Overview of language models)		RW-3 AV-5	Lecture 30 will be utilized to discuss the fundamentals of prompts, the importance of prompt engineering, an overview of language models, and the key elements of a good prompt.	Students will learn about the fundamentals of prompts, the importance of prompt engineering, an overview of language models, and the key elements of a good prompt.	Powerpoint presentation, Demonstration using images and videos, class room discussion	Bad Prompt vs. Good Prompt Challenge

Week 10	Lecture 30	Prompt Engineering(Key elements of a good prompt)		RW-3 AV-5	Lecture 30 will be utilized to discuss the fundamentals of prompts, the importance of prompt engineering, an overview of language models, and the key elements of a good prompt.	Students will learn about the fundamentals of prompts, the importance of prompt engineering, an overview of language models, and the key elements of a good prompt.	Powerpoint presentation, Demonstration using images and videos, class room discussion	Bad Prompt vs. Good Prompt Challenge
Week 11	Lecture 31	Prompt Engineering(Prompt Patterns)		RW-3	Lecture 31 will be utilized to discuss prompt patterns and prompt tuning techniques.	Students will learn about prompt patterns and prompt tuning techniques.	Powerpoint presentation, Demonstration using images, and class room discussion	Bad Prompt vs. Good Prompt Challenge
		Prompt Engineering(Prompt Tuning)		RW-3	Lecture 31 will be utilized to discuss prompt patterns and prompt tuning techniques.	Students will learn about prompt patterns and prompt tuning techniques.	Powerpoint presentation, Demonstration using images, and class room discussion	Bad Prompt vs. Good Prompt Challenge
	Lecture 32				Test 2			
	Lecture 33	Data Analysis & Visualization(Data analysis and visualization using AI tools (ChatGPT Advanced Data Analysis, Tableau))		AV-7	Lecture 33 will be utilized to discuss data analysis and visualization using AI tools.	Students will learn about data analysis and visualization using AI tools.	PowerPoint presentation, Live tool demonstration	
Week 12	Lecture 34				Skill Based Assignment			
	Lecture 35	Data Analysis & Visualization(Working with structured and unstructured data)		AV-7	Lecture 35 will be utilized to discuss working with structured and unstructured data, and an introduction to data pipelines and automation.	Students will learn about working with structured and unstructured data, and an introduction to data pipelines and automation.	PowerPoint presentation, Whiteboard	
		Data Analysis & Visualization(Data pipelines and automation)		AV-7	Lecture 35 will be utilized to discuss working with structured and unstructured data, and an introduction to data pipelines and automation.	Students will learn about working with structured and unstructured data, and an introduction to data pipelines and automation.	PowerPoint presentation, Whiteboard	

Week 12	Lecture 36	AI Model Environments & Lifecycle Basics(Cloud services)		AV-7	Lecture 36 will be utilized to discuss cloud services and edge deployment for AI applications.	Students will learn about cloud services and edge deployment for AI applications.	Powerpoint presentation, Demonstration using images, and classroom discussion	
		AI Model Environments & Lifecycle Basics(Edge deployment)		AV-7	Lecture 36 will be utilized to discuss cloud services and edge deployment for AI applications.	Students will learn about cloud services and edge deployment for AI applications.	Powerpoint presentation, Demonstration using images, and classroom discussion	
Week 13	Lecture 37	AI Model Environments & Lifecycle Basics(Error identification)		AV-7	Lecture 37 will be utilized to discuss error identification and troubleshooting in AI systems.	Students will learn about error identification and troubleshooting in AI systems.	PowerPoint presentation, Whiteboard, Case-based examples	
		AI Model Environments & Lifecycle Basics (Troubleshooting)		AV-7	Lecture 37 will be utilized to discuss error identification and troubleshooting in AI systems.	Students will learn about error identification and troubleshooting in AI systems.	PowerPoint presentation, Whiteboard, Case-based examples	
	Lecture 38	AI Model Environments & Lifecycle Basics(AI process automation)		AV-7	Lecture 38 will be utilized to discuss AI process automation and its applications.	Students will learn about AI process automation and its applications.	PowerPoint presentation, Whiteboard	
	Lecture 39	AI Model Environments & Lifecycle Basics (Introduction to MLOps and lifecycle management)		AV-7	Lecture 39 to 40 will be utilized to discuss the introduction to MLOps and machine learning lifecycle management.	Students will learn about the introduction to MLOps and machine learning lifecycle management.	Powerpoint presentation, Demonstration using images	
Week 14	Lecture 40	AI Model Environments & Lifecycle Basics (Introduction to MLOps and lifecycle management)		AV-7	Lecture 39 to 40 will be utilized to discuss the introduction to MLOps and machine learning lifecycle management.	Students will learn about the introduction to MLOps and machine learning lifecycle management.	Powerpoint presentation, Demonstration using images	
SPILL OVER								
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Scheme for CA:

CA Category of this Course Code is:A0304 (3 best out of 4)

Component	Weightage (%)	Mapped CO(s)
Recoup CA		
Skill Based Assignment	33.33	CO1, CO2, CO3, CO4, CO5, CO6
Test 1	33.33	CO1, CO2
Test 2	33.33	CO3, CO4, CO5

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Skill Based Assignment	This assessment tests students ability to solve real-world problems through innovative AI products or patentable solutions, with focus on technical skills, creativity, and societal impact.	Students are required to create a small AI product, such as a chatbot, utilizing modern Generative AI tools, or draft a patent showcasing a novel AI-driven solution to a real-world problem. Marks distribution for students would be: Problem Identification & Innovation- 10, Technical Execution & Skill Demonstration- 10, and Presentation and Q&A- 10.	Group	Offline	30	4 / 12
Test 1	To check the learnability of the students.	Syllabus of the test will be unit 1 and unit 2.	Individual	Offline	30	5 / 6
Test 2	To check the learnability of the students.	Syllabus of the test will be unit 3, unit 4 and unit 5.	Individual	Offline	30	10 / 11
Recoup CA	subjective test based on the syllabus	Additional chance of CA	Individual	Online	30	8 / 10

Detailed Plan For Practicals

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	List of Practicals	Implement a simple AI problem as a State-Space Search (e.g., Water Jug or 8-Puzzle).		Student will learn to implement state-space search techniques to solve simple AI problems.
Practical 2	List of Practicals	Explore Modern AI Toolkits (TensorFlow, PyTorch): Create a basic AI workflow to classify simple data.		Student will learn to use modern AI toolkits to build a basic AI workflow for data classification.
Practical 3	List of Practicals	Implement Best-First Search and A* algorithms for pathfinding or optimization.	AV-2	Student will learn to implement Best-First Search and A* for problem solving.
Practical 4	List of Practicals	Solve a Constraint Satisfaction Problem (e.g., N-Queens or Sudoku Solver).	RW-1	Student will learn to model and solve constraint satisfaction problems using AI techniques.
Practical 5	List of Practicals	Build and evaluate a Supervised Learning Model (classification or regression) using scikit-learn.		Student will learn to build, train, and evaluate supervised learning models using standard machine learning tools.
Practical 6	List of Practicals	Construct a Bayesian Network and perform inference for a real-life scenario.		Student will learn to construct Bayesian networks and perform probabilistic inference for decision making.
Practical 7	List of Practicals	Implement a Neural Network (CNN) for image classification using TensorFlow or PyTorch.		Student will learn to design and implement convolutional neural networks for image classification tasks.
Practical 8	List of Practicals	Develop a Simple Chatbot using pre-trained Transformer or GPT-based API.		Student will learn to develop a basic conversational AI system using API.
Practical 9	List of Practicals	Experiment with Generative AI Models (e.g., text or image generation using GPT/DALL·E).		Student will learn to use generative AI models for content generation tasks.
Practical 10	List of Practicals	Apply Prompt Engineering Techniques (few-shot, role-based, chain-of-thought) to improve model responses.		Student will learn to apply prompt engineering techniques to enhance the quality of AI model outputs.
Practical 11	List of Practicals	Perform Data Analysis and Visualization using ChatGPT Advanced Data Analysis.		Student will learn to analyze datasets and generate visual insights using AI-assisted data analysis tools.
	SPILL OVER			
Practical 12	Spill Over			